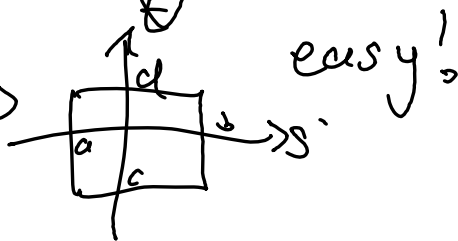
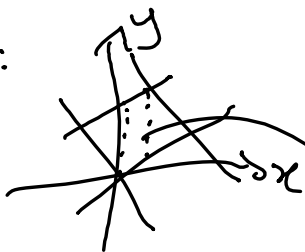


Lecture 24: Change of Variables for double integrals triple integrals, read your memes! Stewart also has good exercises

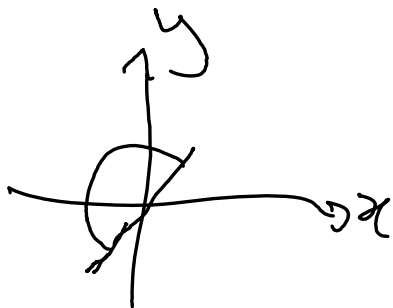
we've already hinted that sometimes changing
our coordinate system is very helpful!

x, y : parallelogram



easy!

$$a \leq s \leq b \\ c \leq t \leq d$$



$$r, \theta: c \leq r \leq a \\ \frac{\pi}{4} \leq \theta \leq \frac{5\pi}{4}$$

Theorem: Let $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous f^n .

Let $T: R \subset \mathbb{R}^2 \rightarrow D$ be a f^n w/ $T(s,t) = (x,y)$, where R is a rectangle! Assume that

① T is invertible (so $T^{-1}(x,y) = (s,t)$)

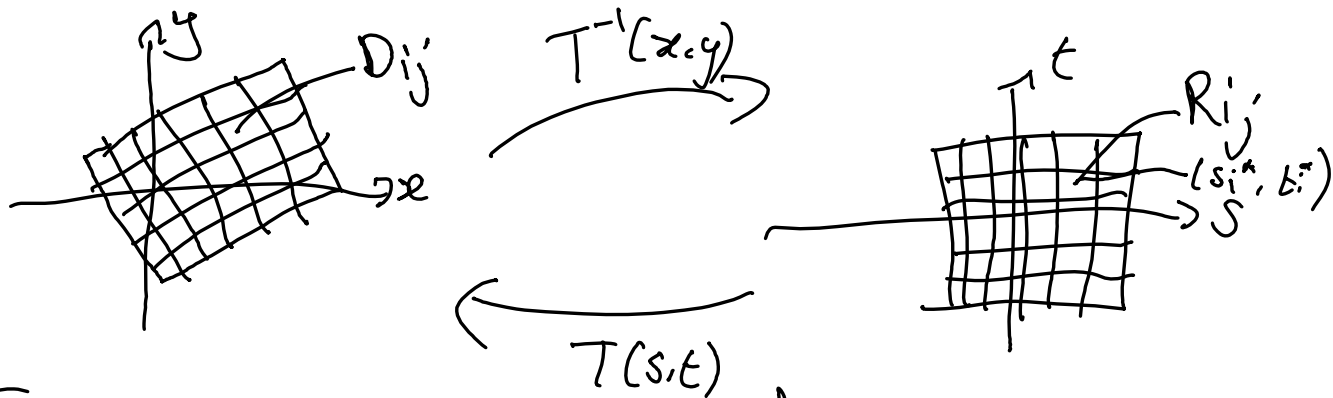
② T and T^{-1} are of class C^1

$$\textcircled{3} J_T(s,t) = \begin{vmatrix} x_s(s,t) & x_t(s,t) \\ y_s(s,t) & y_t(s,t) \end{vmatrix} \neq 0$$

Then

$$\iint_D f(x,y) dx dy = \iint_R f(x(s,t), y(s,t)) |J_T(s,t)| ds dt$$

J_T is called the Jacobian of the transformation T

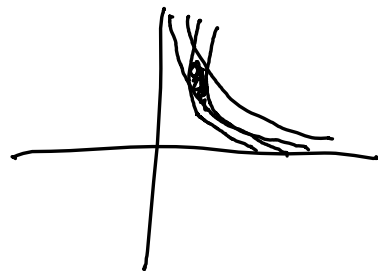


J_T represents the scalar factor between the area of each subrectangle R_{ij} and the area of the corresponding patch D_{ij} in D through the transform. T . thus

$$A(D_{ij}) = |J_T(s_i^*, t_i^*)| R_{ij}$$

Example: $\iint_D xy \, dx \, dy$ where D is enclosed by

$$y = \frac{1}{x}, \quad y = \frac{2}{x}, \quad y = \frac{1}{x^2}, \quad y = \frac{2}{x^2}$$



want to write $nr \leq x,y \leq nr$
 $nr \leq x,y \leq nr$

since $\frac{1}{x} < \frac{2}{x}$ and $\frac{1}{x^2} < \frac{2}{x^2}$, I'm going to
guess $\frac{1}{x} \leq y \leq \frac{2}{x}$ and $\frac{1}{x^2} \leq y \leq \frac{2}{x^2}$

so $1 \leq xy \leq 2$ and $1 \leq x^2 y \leq 2$

now, let $s = xy$, $t = x^2 y$ now I want $\begin{matrix} s, t \\ x = \dots \\ y = \dots \end{matrix}$

$$s = xy, \quad t = x^2 y$$

try make s and t interact to get just x or just y out! This is the tricky part!

notice $\frac{t}{s} = \frac{x^2 y}{xy} = x$

notice $s^2 = x^2 y^2$ so $\frac{s^2}{t} = \frac{x^2 y^2}{x^2 y} = y$ so

$x = \frac{t}{s}, \quad y = \frac{s^2}{t},$ so our new domain is

$$1 \leq s \leq 2, \quad 1 \leq t \leq 2$$

$$J = \begin{vmatrix} x_s & x_t \\ y_s & y_t \end{vmatrix} = \begin{vmatrix} -\frac{t}{s^2} & \frac{1}{s} \\ \frac{2s}{t} & -\frac{s^2}{t^2} \end{vmatrix} = \frac{t s^2}{s^2 t^2} - \frac{2s}{s t} = \frac{1}{t} - \frac{2}{t} = -\frac{1}{t} \quad \text{so } |J| = \frac{1}{t}$$

let's integrate!

$$\iint_D xy \, dx \, dy$$

$$f(x, y) = xy$$

$$f\left(\frac{t}{s}, \frac{s^2}{t}\right) = s$$

$$\int_1^2 \int_1^2 s \cdot \frac{1}{t} \, ds \, dt = \int_1^2 s \, ds \int_1^2 \frac{1}{t} \, dt$$

$$= \left[\frac{s^2}{2} \right]_1^2 \cdot \left[\ln t \right]_1^2 = \left(\frac{4}{2} - \frac{1}{2} \right) (\ln 2 - \ln 1) = \frac{3}{2} \ln 2$$

But for any rectangular integral, if $f(x, y) = f^x(x) \cdot f^y(y)$ then you can turn it into a product

$$\int_c^d \int_a^b f(x, y) \, dx \, dy = \int_a^b f^x(x) \, dx \int_c^d f^y(y) \, dy$$

e.g. from tut

$$\int_0^b \int_c^d x \arctan \frac{1}{x} \cos(y^2) \, dy \, dx = \int_a^b dx \int_c^d dy$$
$$= \int_a^b x \arctan \frac{1}{x} \, dx \int_c^d \cos(y^2) \, dy$$

something like $\iint \sin(x-y) \, dx \, dy$ you cannot split like this!

we've already seen a polar example. let me
just convince you $|J_{\text{polar}}| = r$

$$T(r, \theta) = (r \cos \theta, r \sin \theta) = (x, y) \quad \begin{array}{l} \text{so } x = r \cos \theta \\ y = r \sin \theta \end{array}$$

$$J_T = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

$$\text{and } |J_T| = |r| = r.$$

one more e.g. $D = \{(x, y) \in \mathbb{R}^2 : -1 \leq 2x+y \leq 1, 0 \leq -x+y \leq 2\}$

$$f(x, y) = 2x - y$$

let $s = 2x + y$ let $t = -x + y$

$$s - t = 2x + y + x - y = 3x \quad \text{so} \quad \frac{s-t}{3} = x$$

$$s + 2t = 2x + y - 2x + 2y = 3y \quad \text{so} \quad \frac{s+2t}{3} = y$$

$$J_T = \begin{vmatrix} x_s & x_t \\ y_s & y_t \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} = \frac{2}{9} + \frac{1}{9} = \frac{1}{3}$$

so $\iint_D f(x, y) \, dx \, dy = \int_0^2 \int_{-1}^1 \left(\frac{s-t}{3} - \left(\frac{s+2t}{3} \right) \right) \cdot \frac{1}{3} \, ds \, dt$

$$= \int_0^2 \int_{-1}^1 \frac{-3t}{9} \, ds \, dt = -\frac{1}{3} \int_0^2 \int_{-1}^1 t \, ds \, dt$$

triple integrals!

let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous f'' and let

$$R = [a_1, a_2] \times [b_1, b_2] \times [c_1, c_2]$$

$$\iiint_R f(x, y, z) dx dy dz = \int_{c_1}^{c_2} \int_{b_1}^{b_2} \int_{a_1}^{a_2} f(x, y, z) dx dy dz$$

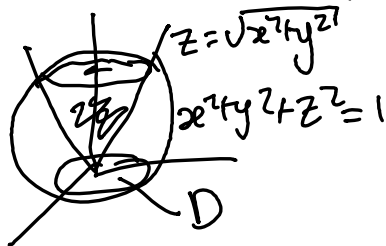
non-rectangular region...

Suppose $R = \{ (x, y, z) \in \mathbb{R}^3 : (x, y) \in D \subset \mathbb{R}^2 \text{ and } g_1(x, y) \leq z \leq g_2(x, y) \}$

where $g_1, g_2 : D \rightarrow \mathbb{R}$ are continuous fns s.t.

$$g_1(x, y) \leq g_2(x, y) \quad \forall (x, y) \in D$$

e.g



$$g_1(x, y) = \sqrt{x^2 + y^2}$$

$$g_2(x, y) = \sqrt{1 - x^2 - y^2}$$

D is the circle where those points exist

Then

$$\iiint_R f(x, y, z) dx dy dz = \iint_D \left(\int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz \right) dx dy$$